535518 Spring Semester 2026: Deep Learning **(Due: 2026/05/05, Tuesday, 23:59)**

Lab 5: Value-Based Reinforcement Learning

Instructor: Ping-Chun Hsieh
TAs: Kai-Siang Ma, Po-Hsuan Wen

Intended Learning Outcomes:

- Understand the core components of the Deep Q-Network algorithm.
- Gain hands-on experience implementing DQNs in PyTorch.
- Learn how to use DQNs in both low-dimensional and high-dimensional (visual) environments.
- Explore several practical enhancements to the DQN algorithm: Double DQN, Prioritized Experience Replay, and multi-step return.

1 Background

In this assignment, you will implement and experiment with Deep Q-Networks (DQNs) to solve classic control and visual reinforcement learning (RL) tasks using PyTorch and OpenAI Gym environments. The homework consists of three main parts, progressing from basic DQN implementation to incorporating advanced techniques to improve learning efficiency.

Vanilla DQN. As described in Lecture 8, DQN [3] combines Q-learning with a neural function approximator and leverages two practical techniques, namely *experience replay* and *target network*. Specifically, experience replay refers to the use of a replay memory that stores transitions (s, a, r, s') collected from the environment, and target network refers to maintaining a second Q network which is updated only periodically every N updates from the main Q network. The training of DQN centers around the minimization of the following loss function

$$L_{\text{DQN}}(\theta) := \frac{1}{2} \sum_{(s,a,r,s')\in D} \Big(r + \gamma \max_{a'\in\mathcal{A}} Q(s', a'; \bar{\theta}) - Q(s, a; \theta) \Big)^2, \tag{1}$$

where D denotes the sampled batch of transitions from the replay memory, and θ and $\bar{\theta}$ denote the network parameters of the main Q network and the target Q network, respectively.

Enhancements to vanilla DQN. To improve the sample efficiency of DQN, various techniques have been proposed [2, 1], such as enhancements for the loss function, the replay memory sampling strategy, and the exploration strategy. In this lab, we focus on the following three techniques:

- **Double DQN (DDQN)**: To mitigate the overestimation bias in DQN, one can apply the double Q-learning technique, which decouples the maximization step in the loss function of DQN into two sub-steps, i.e., using the main Q network for selecting the greedy action while using the target Q network for taking the Q values. Accordingly, the loss function of DDQN is

$$L_{\text{DDQN}}(\theta) := \frac{1}{2} \sum_{(s,a,r,s')\sim D} \Big(r + \gamma Q\big(s', \arg\max_{a'\in A} Q(s', a'; \theta); \bar{\theta}\big) - Q(s, a; \theta) \Big)^2 \tag{2}$$

- **Prioritized Experience Replay (PER)**: PER [4] improves learning by sampling transitions with higher learning potential more frequently. Each transition i in the replay buffer is assigned a priority p_i, often based on the magnitude of its Bellman error:

$$p_i = |\delta_i| + \epsilon$$

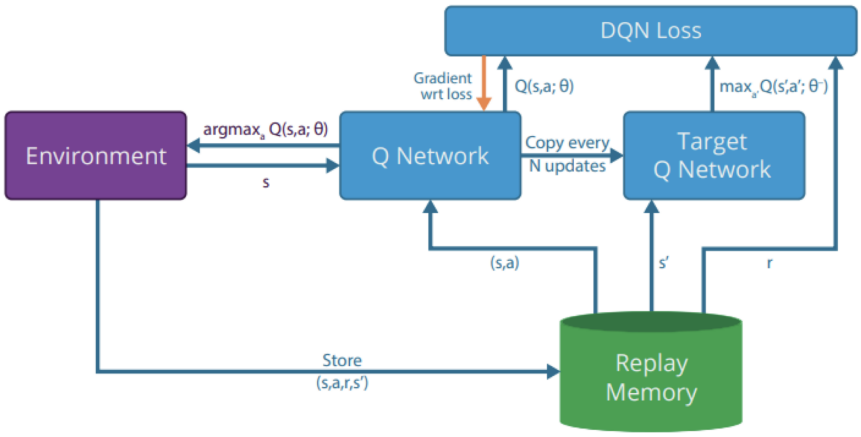


Figure 1: The architecture of vanilla DQN.

where $\delta_i = r_i + \gamma \max_{a'} Q(s'_i, a') - Q(s_i, a_i)$ is the Bellman error, and $\epsilon > 0$ is a small constant to ensure all transitions are sampled with non-zero probability. The probability of sampling transition i is:

$$P(i) = \frac{p_i^\alpha}{\sum_k p_k^\alpha}$$

where $\alpha \in [0, 1]$ controls the degree of prioritization. To correct the introduced bias, importance-sampling (IS) weights are used during training:

$$w_i = \left(\frac{1}{N \cdot P(i)} \right)^\beta$$

where N is the replay buffer size and $\beta \in [0, 1]$ controls the degree of correction. These weights are typically normalized to stabilize learning.

- **Multi-Step Return**: Multi-step learning in Deep DQN enhances learning efficiency by propagating rewards over multiple time steps, allowing the agent to learn from longer-term outcomes. Instead of using the immediate reward for a single transition, multi-step DQN uses the accumulated return over n steps. The n-step return $R_t^{(n)}$ is defined as:

$$R_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n \max_{a'} Q(s_{t+n}, a')$$

Here, r_{t+k} is the reward at time $t + k$, γ is the discount factor, and $Q(s_{t+n}, a')$ is the estimated future return from the state s_{t+n}. This return replaces the target in the loss function, helping the agent learn more effectively from delayed rewards and reducing the bias-variance tradeoff in value estimation.

2 Lab Description

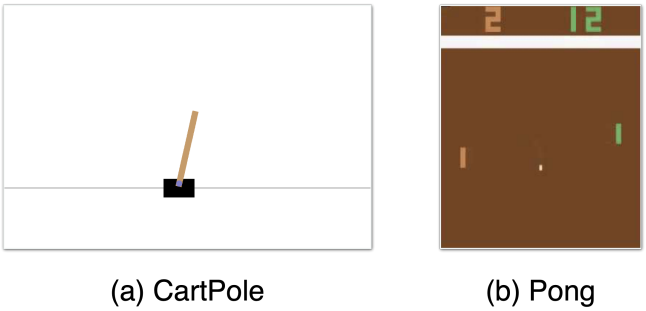
2.1 Task 1: Vanilla DQN in a toy environment

Goal: Implement a vanilla DQN using PyTorch to solve the `CartPole-v1` (https://gymnasium.farama.org/environments/classic_control/cart_pole/) from OpenAI Gym.

Requirements:

- Use `dqn.py` as the starting point for your implementation.

- Use a simple fully connected neural network to approximate the Q-function. The model size can be configured freely by yourself.
- Implement an epsilon-greedy policy for action selection.
- Use experience replay with uniform sampling and a target network.
- Evaluate and plot the total episodic rewards versus environment steps (preferably via Weight and Bias).



(a) CartPole (b) Pong

Figure 2: The evaluation domains used in Lab 5.

2.2 Task 2: Vanilla DQN with Visual Observations on Atari

Goal: Extend your DQN implementation to work on high-dimensional visual input using the `Pong-v5` environment from the Arcade Learning Environment suite, also known as Atari. You can find the detailed definitions of the states, actions, and rewards of each environment at the official website https://ale.farama.org/index.html.

Requirements:

- Preprocess the input frames (grayscale, resize, and stack frames)
- Use a convolutional neural network (CNN) as the Q-function approximator
- Evaluate and plot the total episodic rewards versus environment steps (preferably via Weight and Bias).

```python
import gymnasium as gym
import ale_py

gym.register_envs(ale_py)

# Initialise the environment
env = gym.make("ALE/Breakout-v5", render_mode="human")

# Reset the environment to generate the first observation
observation, info = env.reset(seed=42)
for _ in range(1000):
    # this is where you would insert your policy
    action = env.action_space.sample()

    # step (transition) through the environment with the action
    # receiving the next observation, reward and if the episode has terminated or truncated
    observation, reward, terminated, truncated, info = env.step(action)

    # If the episode has ended then we can reset to start a new episode
    if terminated or truncated:
        observation, info = env.reset()

env.close()
```

Figure 3: The typical usage of the Atari environments in RL.

2.3 Task 3: Enhanced DQN

Goal: Improve the learning efficiency of your DQN agent by incorporating the following enhancements:

- Double DQN
- Prioritized Experience Replay (PER)
- Multi-Step Return

Requirements:

- Integrate the enhancements into your DQN code (you may start from either CartPole or Pong implementation).
- Justify the integration choices.
- Compare training performance (e.g., learning speed, final reward) against vanilla DQN using the `Pong-v5` environment.
- Mark the number of timesteps the model reaches score 19 (the first time after convergence) in your report. We will rerun your model snapshots and judge accordingly if you fail to justify the timestep you reached 19.

3 Model and Package Configurations

- All the modules or classes (listed below) are already provided in the file `dqn.py`. Please do not change the structure of the modules.
  - DQN
  - PrioritizedReplayBuffer
  - AtariPreprocessor
  - DQNAgent
- The model sizes can be freely configured by yourself. That being said, it is expected that Q networks with only a few hidden layers (less than 8) would be sufficient for successful training.
- Regarding the GPU memory usage, you may need to pay attention to the size of your replay memory, especially for the Atari game. Typically, the memory size is set to around 50k to 500k, depending on your GPU memory.
- Training in Pong takes much longer than that in CartPole. As a rule of thumb, Pong typically requires about up to 1 million environmental steps before converging to a dominant score of around 21 points. This could convert to up to about 20 hours of wall clock time (e.g., under a RTX3090 GPU). **Therefore, please do start early so that you have sufficient time for training**.
- The recommended package versions are as follows:
  - `Python` >= 3.8
  - `gymnasium` 1.1.1
  - `ale-py` >= 0.10.0
  - `opencv-python`
  - `torch`
  - `wandb`

4 Grading Policy

4.1 Report (up to 60%)

- Introduction (5%): Please provide a high-level introduction to your report. You can mention **the most important findings and the overall organization** of this report.
- Your implementation (20%): Please briefly explain both the concepts and your implementation details for Tasks 1–3. For each item below, you should first describe how the method works (e.g., how to compute Bellman error) and then explain how you implemented it in your code. Specifically, please describe:
  - How do you obtain the Bellman error for DQN?
  - How do you modify DQN to Double DQN?
  - How do you implement the memory buffer for PER?
  - How do you modify the 1-step return to multi-step return?
  - Explain how you use Weight & Bias to track the model performance.
- Analysis and discussions (25%)
  - Plot the training curves (evaluation score versus environment steps) for Task 1, Task 2, and Task 3 separately, please make sure the **x-axis represent environment step** (3+3+4=10%).
  - Analyze the sample efficiency with and without the DQN enhancements (10%).
  - Please perform an ablation study on each technique separately (5%).
- Additional analysis on other training strategies (Bonus up to 10%)

4.2 Attained Model Performance: Snapshots and Demo Video (up to 50%)

4.2.1 Demo Video

Please record a demo video that showcases two things via screen recording:

- Your source code: Please use about 2 minute to describe your implementation.
- Your model performance: Please use about 3 minute to demo your obtained models for Task 1, Task 2, and Task 3.

The demo video shall be between 5-minute to 6-minute long. **Please prepare your demo video in English** (unless there is any special circumstances discussed beforehand with the TAs). **Notably, your model snapshots will NOT be graded if you do not provide a valid demo video.**

4.2.2 Model Snapshots

Grading Criteria:

- Task 1 (15%): The grading of Task 1 would depend on the evaluation score of your submitted snapshot. **Please use the best snapshot that you have obtained during the training process.** The score will be determined by the following equation

$$\text{Score percentage} = \frac{\min\{\text{average evaluation score}, 480\}}{480} \times 15\% \tag{3}$$

  Accordingly, to get full score for this part, your vanilla DQN shall constantly achieve an average score above 480 (over 20 evaluation episodes).

- Task 2 (20%): Similarly, the grading of Task 2 would depend on the evaluation score of your submitted snapshot. **Please use the best snapshot that you have obtained during the training process.** The score will be determined by the following equation

$$\text{Score percentage} = \frac{\min\{\text{average evaluation score}, 19\} + 21}{40} \times 20\% \tag{4}$$

  Accordingly, to get full score for this part, your DQN shall steadily achieve an average score above 19 (over 20 evaluation episodes).

- Task 3 (20%): The grading of Task 3 would depend on the **sample efficiency of your enhanced DQN**. Please submit 5 model snapshots that are trained for 600k, 1M, 1.5M, 2M, and 2.5M environment steps.

Environment steps needed (Reaching score 19 on Pong)	600k	1M	1.5M	2M	2.5M	>2.5M
Score Percentage	20%	17%	15%	13%	11%	8%

Evaluation Protocol: To ensure the reproducibility of your results, please adhere to the following evaluation protocol:

- **Provide a runnable command or script:** (e.g., a `.sh` file or a command-line instruction) which can be directly executed by the TAs to evaluate your submitted snapshot. Make sure that all necessary arguments, paths, and environment variables are clearly specified.
- **Provide your environment requirements:** To ensure the TAs could reproduce your results, please attach the ‘requirements.txt‘ in your submission.
- **Provide the screenshots of your evaluation results:** Please include the screenshots of your evaluation results (as demonstrated in Figure 4) for each task in your report and the mean reward of the 20 evaluation episodes. A 3-point penalty will be deducted (from the Report 60%) for *each* missing screenshot.
- **Submit the model snapshots:** Provide the exact model files those were used to achieve your reported performance.
- **Important:** Please include the execution command for each task, if the TAs cannot reproduce your results, a 5-point penalty will be applied.
- **Example command:** `python test_model_task3.py --model_path LAB5_StudentID_task3_2500000.pt`

  Evaluation seeds: 0 to 19

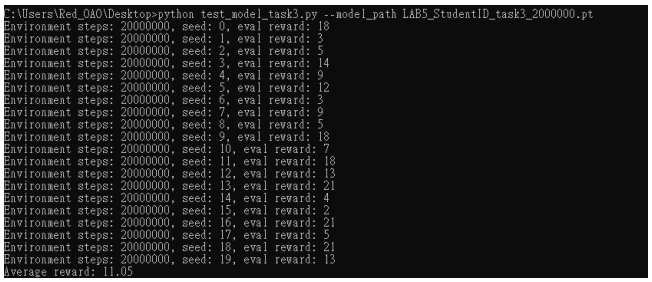


Figure 4: Example of the evaluation of task3 in 2M environment steps.

5 Format of Submitted Files

You should submit a single ZIP file that contains all your deliverables. Please strictly follow the naming and structure below:

- **ZIP file**: Compress all files into one archive.

- **Filename**: `LAB5_{StudentID}_{YourName}.zip`

The ZIP file should contain the following structure:

Directory Structure

```
LAB5_StudentID.zip
|-- LAB5_StudentID_Code/         <- Source code folder
|   |-- (any other .py files)    <- Your code files
|   |-- requirements.txt
|   |-- (any other .sh files)    <- Optional
|-- LAB5_StudentID.pdf           <- Technical report (single PDF)
|-- LAB5_StudentID.mp4           <- Demo video (5 - 6 minutes)
|-- LAB5_StudentID_task1.pt      <- Task 1 model snapshot
|-- LAB5_StudentID_task2.pt          <- Task 2 model snapshot
|-- LAB5_StudentID_task3_600000.pt    <- Task 3 snapshot (step = 600k)
|-- LAB5_StudentID_task3_1000000.pt    <- Task 3 snapshot (step = 1M)
|-- ...
|-- LAB5_StudentID_task3_2500000.pt   <- Task 3 snapshot (step = 2.5M)
|-- LAB5_StudentID_task3_best.pt   <- Task 3 snapshot (any step reach score 19)
```

Note:

- Only source code files should go into the `Code` folder.
- All other items (report, video, models) must be placed directly in the ZIP root.
- A 5-point penalty will be applied for any incorrect filenames or folder structure.
- Important Dates:
  - Code & report submission deadline: 05/05 (Tue.) 23:59
  - Submissions made after the deadline but before 05/21 (Thu.) 23:59 will still be accepted, but the final score will be multiplied by 0.7 (i.e., a 30% discount on the final score of this homework).
  - Submissions after 5/21 23:59 will not be accepted and will receive a grade of 0.

References

[1] J. S. O. Ceron and P. S. Castro. Revisiting Rainbow: Promoting more insightful and inclusive deep reinforcement learning research. In *International Conference on Machine Learning*, pages 1373–1383, 2021.

[2] M. Hessel, J. Modayil, H. Van Hasselt, T. Schaul, G. Ostrovski, W. Dabney, D. Horgan, B. Piot, M. Azar, and D. Silver. Rainbow: Combining improvements in deep reinforcement learning. In *Proceedings of the AAAI conference on artificial intelligence*, volume 32, 2018.

[3] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, et al. Human-level control through deep reinforcement learning. *Nature*, 518 (7540):529–533, 2015.

[4] T. Schaul, J. Quan, I. Antonoglou, and D. Silver. Prioritized experience replay. In *International Conference on Learning Representations*, 2016.